

User Guide to kotlin-program.

May 11, 2026.

1. Introduction.

Let k , p and n be integers such that $0 \leq k \leq k + p < n$. The symmetric shift register $\theta : \{0, 1\}^n \rightarrow \{0, 1\}^n$ with respect to k , p and n is defined by $\theta(a_1 \cdots a_n) = a_2 \cdots a_{n+1}$ where

$a_{n+1} = a'_1$ if $k \leq a_2 + \cdots + a_n \leq k + p$, and $a_{n+1} = a_1$ otherwise, where $a'_1 = 0$ if $a_1 = 1$, and $a'_1 = 1$ if $a_1 = 0$.

The kotlin-program determine minimal periods of bit strings generated by symmetric shift registers in the main case described in Section 6. The periods are calculated by inductive formulas described in the article "Representation of symmetric shift registers". In this paper we also describe how every case can be reduced to the main case. In Section 2 we describe the input and some limitations. The remaining sections are divided into 10 parts.

Part 1 describe assumptions, notation and the main case.

In Part 2 we describe the output of the program. In addition to the minimal periods the output gives an algorithmic description of how the periods are calculated. Moreover, a lot of intermediate results are displayed. We illustrate this by examples.

In Part 3 we explain how every case can be reduced to the main case.

The calculations and the formulation of the results depend on parameters and vectors that are defined and exemplified in Part 4, $\cdots$, Part 9.

In Part 10 we deduce the output results of the examples in Part 2.

"Representation of symmetric shift registers" by Jan Sørensen is published and available at arXiv [2505.23974].

The kotlin-program is made by Pål, Eirik og Martin Sørensen.

2. Input to the kotlin-program.

Input: $Q = (v_1, \dots, v_s)$ where s is even, $v_i > 0$ for $1 \leq i \leq s - 1$ and $v_s \geq 0$.

The input vector is comma separated and must be given without parentheses. After the vector is entered, an admissible p -value must be chosen. Valid range is specified. Next, click on "Compute period". Then the periods and intermediate results are displayed.

It is also possible to check directly if the calculated periods are correct by using the definition of the shift register step by step. However, this can only be done if the periods are not too large.

The admissible p -values of Q are defined in Section 11. In "Representation of symmetric shift registers" it is described how every case can be reduced to the case where we use an admissible p -value of Q .

The program works if p and the coordinates of Q are not too large. These restrictions are due to the fact that the values of the integer variables must remain within certain limits. It is difficult to predict the size of the calculated periods. Therefore it is not easy to say anything precisely about how large values Q and p can have.

In the next parts we will interpret the input and the output results. Here follows an interpretation on an example. We need some notation.

If $j \geq 1$, then 0_j denotes j consecutive 0's, and 1_j denotes j consecutive 1's. Moreover, if $j = 0$, then 0_j and 1_j denote the empty string.

If $A = a_1 \dots a_n$ is a string, then

$$w(A) = \text{the number of bits equal to 1 in } A = a_1 + \dots + a_n.$$

Example 2.1. Suppose $Q = (3, 2, 1, 4)$. In this case the input vector is given as 3,2,1,4 (the parentheses are removed). The vector Q corresponds to

$$A = 1_3 0_2 1_1 0_4 = 1110010000.$$

In this case the admissible p -values are 0, 1 and 2. Suppose we choose $p = 2$. Let $k = w(A) - (p + 1) = 4 - (2 + 1) = 1$ and $n = \text{length}(A) = 10$.

Let A^∞ be generated from A by the symmetric shift register with respect to k, p and n . We call A^∞ the shift symmetric bit string generated by Q and p . By applying the kotlin-program we get that $\zeta = 95$ is the minimal period of A^∞ , and $j = 32$ is the least even vector period of the corresponding vector representation. The vector representation is defined in Section 4.

Table of contents in the remaining parts.

Part 1.

- Section 3. Assumptions.
- Section 4. Infinite vector representations.
- Section 5. Representation by vectors.
- Section 6. The main case.

Part 2.

- Section 7. Output of the kotlin-program.
- Section 8. Examples of output.
- Section 9. Interpretation of the results.

Part 3.

- Section 10. The reduction to the main case.

Part 4.

- Section 11. Admissible p -values.
- Section 12. Admissible p -values - an algorithmic approach.

Part 5.

- Section 13. The distance measure.
- Section 14. The distance function.
- Section 15. Examples.

Part 6.

- Section 16. Auxiliary functions.
- Section 17. The r -indexes.

- Section 18. The contraction vector.

Part 7.

- Section 19. The extension of vectors.
- Section 20. Cyclic parameters.

Part 8.

- Section 21. The parameter α .
- Section 22. The distance vector.

Part 9.

- Section 23. Progression parameters.

Part 10.

The output results in Example 8.1, 8.2 and 8.3 are deduced.

Part 1.

We describe in this part the assumptions and the type of problems that can be solved by the kotlin-program.

3. Assumptions.

We suppose $A = a_1 \cdots a_n$ where a_i are bits for $1 \leq i \leq n$, $k \geq 0$ and $p \geq 0$ such that $0 \leq k < k + p + 1 \leq n$. Let $w(A) = a_1 + \cdots + a_n$ = the number of ones in A . We also suppose $k \leq w(A) \leq k + p + 1$.

We suppose $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ is the bit string generated from A by the symmetric shift register with parameters k , p and n . For $i \geq 0$, then (3.1) $a_{n+i+1} = a'_{i+1}$ if $k \leq a_{i+2} + \cdots + a_{i+n} \leq k + p$. Otherwise, $a_{n+i+1} = a_{i+1}$. We note that $a' = 0$ if $a = 1$, and $a' = 1$ if $a = 0$.

4. Infinite vector representations.

Let $A^\infty = a_1 a_2 a_3 \cdots$ be generated from $A = a_1 \cdots a_n$ by the symmetric shift register with parameters k , p and n where A starts with 1.

As previously, 0_j denotes j consecutive 0's and 1_j denotes j consecutive 1's. If $j = 0$, then 0_j and 1_j denote the empty string. It can be proved that A^∞ can be decomposed as

$$(4.1) \quad A^\infty = 1_{q_1} 0_{q_2} 1_{q_3} 0_{q_4} \cdots \text{ where } q_i > 0 \text{ for } i \geq 1.$$

Then $V(A^\infty) = (q_1, q_2, \cdots)$ is called the vector representation of A^∞ . Next, we suppose $w(A) = k + p + 1$. Then it can be proved that

$$(4.2) \quad V(A^\infty) \text{ has an even vector period,}$$

$$(4.3) \quad q_1 + \cdots + q_j \text{ is the minimal period of } A^\infty \text{ if } j \text{ is the least even vector period of } V(A^\infty).$$

5. Representation by vectors.

Suppose $A = a_1 \cdots a_n$ where $a_1 = 1$. Then A has an even decomposition of the form $A = 1_{v_1} 0_{v_2} 1_{v_3} 0_{v_4} \cdots 1_{v_J} 0_{v_{J+1}}$ where $v_i > 0$ for $1 \leq i \leq J$, $v_{J+1} \geq 0$ and $J \geq 1$ is odd. We call the vector $V(A) = (v_1, \cdots, v_J, v_{J+1})$ the even vector representation of A .

Example 5.1. If $A = 110011100$, then we decompose $A = 1_2 0_2 1_3 0_2$. In this case $V(A) = (2, 2, 3, 2)$ is the even vector representation of A .

Example 5.2. If $A = 111100111$, then we decompose $A = 1_4 0_2 1_3 0_0$. In this case $V(A) = (4, 2, 3, 0)$ is the even vector representation of A .

Suppose $Q = (v_1, \cdots, v_{J+1})$ where $v_i > 0$ for $1 \leq i \leq J$, $v_{J+1} \geq 0$ and $J \geq 1$ is odd. Let $A(Q) = 1_{v_1} 0_{v_2} 1_{v_3} \cdots 1_{v_J} 0_{v_{J+1}}$. We note that $A(Q)$ starts with 1.

Example 5.3. If $Q = (3, 4, 2, 0)$, then $A(Q) = 1_3 0_4 1_2 0_0 = 111000011$.

Example 5.4. If $Q = (2, 3, 1, 3)$, then $A(Q) = 1_2 0_3 1_1 0_3 = 110001000$.

6. The main case.

Suppose $Q = (v_1, \dots, v_s)$ is an even vector such that

$$v_i > 0 \text{ for } 1 \leq i < s, v_s \geq 0 \text{ and } s \text{ is even.}$$

Let $p \geq 0$ be an admissible p -value of Q . The admissible p -values are defined in Section 11. Let

$$A = A(Q) = 1_{v_1} 0_{v_2} \dots 1_{v_{s-1}} 0_{v_s}, k = w(A) - (p + 1) \text{ and } n = \text{length}(A).$$

Let A^∞ be generated from A by the symmetric shift register with respect to k , p and n . Then $w(A) = k + p + 1$ and A starts with 1. Since p is an admissible p -value of Q , it can be proved that $k \geq 0$. We call A^∞ the shift symmetric bit string generated by Q and p .

By using Q and p as input the kotlin-program calculates the minimal period ζ of A^∞ and the least even vector period j of $V(A^\infty)$. That means, ζ is the minimal period of the shift symmetric bit string generated by Q and p , and j is the least even vector period of the corresponding vector representation.

In "Representation of symmetric shift registers" it is described how every case can be reduced to the case where we use an admissible p -value of Q .

Example 6.1. Suppose $Q = (3, 4, 2, 4, 1, 0)$ and $p = 2$. Then p is an admissible p -value of Q . We refer to Observation 11.3. By applying the kotlin-program we get that $\zeta = 982$ is the minimal period of the shift symmetric bit string A^∞ generated by Q and p , and $j = 370$ is the least even vector period of the corresponding vector representation.

In this case A^∞ is generated from $A = A(Q) = 1_3 0_4 1_2 0_4 1_1 0_0 = 11100001100001$ by the symmetric shift register with respect to

$$k = w(A) - (2 + 1) = 3, p = 2 \text{ and } n = \text{length}(A) = 14.$$

Example 6.2. Suppose $Q = (2, 3, 1, 3)$ and $p = 1$. Then p is an admissible p -value of Q . We refer to Observation 11.3. By applying the kotlin-program we get that $\zeta = 68$ is the minimal period of the shift symmetric bit string A^∞ generated by Q and p , and $j = 26$ is the least even vector period of the corresponding vector representation. In this case A^∞ is generated from

$$A = A(Q) = 1_2 0_3 1_1 0_3 = 110001000$$

by the symmetric shift register with respect to $k = w(A) - (1 + 1) = 1$, $p = 1$ and $n = \text{length}(A) = 9$.

Example 6.3. Suppose $Q = (1, 5)$ and $p = 0$. Then p is an admissible p -value of Q . We refer to Observation 11.3. By applying the kotlin-program we get that $\zeta = 7$ is the minimal period of the shift symmetric bit string A^∞ generated by Q and p , and $j = 2$ is the least even vector period of the corresponding vector representation.

In this case A^∞ is generated from $A = A(Q) = 1_1 0_5 = 100000$ by the symmetric shift register with respect to

$$k = w(A) - (0 + 1) = 0, p = 0 \text{ and } n = \text{length}(A) = 6.$$

Part 2.

7. Ouput of the kotlin-program.

Input: $Q = (v_1, \dots, v_s)$ where s is even, $v_i > 0$ for $1 \leq i \leq s - 1$ and $v_s \geq 0$. Suppose we have written an input vector, chosen an admissible p -values of Q and started the program. Then the program produce an output with both the main results and also some of the intermediate results.

The output can be divided into 5 parts. The notation and the content in the output are taken from "Representation of symmetric shift registers". However, there are some differences in the notation. For example, $Q[i]$ is used instead of Q_i , $\alpha[i]$ is used instead of α_i , etc. In Section 8 the calculations of the output are described by examples.

If the periods are not too large, then a simulation-based check of the results are displayed. This is done by using the definition of the symmetric shift register step by step.

Start values: The input vector Q and the p -value are presented.

Main result: The minimal period ζ of the shift symmetric bit string generated by Q and p , and the least even vector period j of the corresponding vector representation.

Contraction results: The vectors $Q[p], Q[p - 1], \dots, Q[0]$ are displayed. These vectors are defined by the contraction map π in Section 18. In fact,

$$Q[p] = Q, Q[p - 1] = \pi(Q[p]), \dots, Q[0] = \pi(Q[1]).$$

Hence, $Q[i - 1]$ is the contraction vector of $Q[i]$ for $1 \leq i \leq p$. In the calculations of $Q[i - 1]$ the τ -vector and r -vector of $Q[i]$ are used. These vectors are also displayed in the output. We refer to the definitions of these vectors in Section 14 and 17 respectively. Moreover, we refer to Section 18 for more details about the calculations of the contraction vectors.

Periodic results:

$$\begin{array}{cccc}
p\text{-value} = p, & Q[p], & j[p], & \zeta[p] \\
p\text{-value} = p-1, & Q[p-1], & j[p-1], & \zeta[p-1] \\
& \vdots & & \\
p\text{-value} = 0, & Q[0], & j[0], & \zeta[0]
\end{array}$$

Suppose $0 \leq i \leq p$. Then $\zeta[i]$ is the minimal period of the shift symmetric bit string generated by $Q[i]$ and the p -value $= i$, and $j[i]$ is the least even vector period of the corresponding vector representation.

Intermediate results: The periodic parameters $j[i]$ and $\zeta[i]$ are calculated by an inductive procedure for $i = 0$ to $i = p$ in $p + 1$ steps as follows.

In Step 0 the parameters $j[0]$ and $\zeta[0]$ are set equal to the cyclic parameters of the extension of $Q[0]$. We refer to Section 19 and 20.

Suppose $j[i-1]$ and $\zeta[i-1]$ are calculated by Step $i-1$ where $0 \leq i < p$. Then $j[i]$ and $\zeta[i]$ are calculated in Step i by dividing into two sub-cases:

Case 1. Suppose $Q[i] = (v_1, \dots, v_s)$ where $v_i > 1$ for $1 \leq i \leq s-1$. Then

$$j[i] = j[i-1] \text{ and } \zeta[i] = \zeta[i-1] + j[i].$$

Case 2. Suppose $Q[i] = (v_1, \dots, v_s)$ has at least one coordinate $v_i = 1$ where $1 \leq i \leq s-1$. Here is the description of the output in this case:

- 1) $C[i]$ is the index vector of $Q[i]$. We refer to Section 22.
- 2) τ is the τ -vector of $Q[i]$. We refer to Section 14.
- 3) $D[i]$ is the distance vector of $Q[i]$. We refer to Section 22.
- 4) $\gamma[i] = \#D[i]$ = the number of coordinates of $D[i]$.
- 5) $\alpha[i] = \delta(Q[i]) + 1$ where δ is the distance measure.
We refer to Section 13 and 21.
- 6) $\alpha^*[i]$ and $\gamma^*[i]$ are the progression parameters of $D[i]$
with respect to $\alpha[i]$. We refer to Section 23.
- 7) $g = \gcd(\alpha^*[i], \zeta[i-1])$ where $\gcd$ means "greatest common divisor".
- 8) Finally, $j[i]$ and $\zeta[i]$ are calculated as follows:

$$\begin{aligned}
x &= \zeta[i-1]/g, & y &= \alpha^*[i]/g \\
j[i] &= 2 \cdot x \cdot \gamma^*[i] + y \cdot j[i-1] \\
\zeta[i] &= y \cdot \zeta[i-1] + j[i].
\end{aligned}$$

We note that $\#$ means "the number of". We refer to Section 8 where the output is described by examples.

8. Examples of output.

In the output from the kotlin-program the notation *alpha*, *zeta* $\dots$ are used. In the description of the following example we use instead α , ζ , $\dots$. In this way we get a smoother presentation. In Part 10 we describe how the output results in this section are deduced.

In Example 8.1 and 8.2 a simulation-based check of the results are displayed. However, in Example 8.3 this is not done since the periods are too large. In the following examples we delete the description of this check.

Example 8.1. Start values: $Q = [4, 4, 2, 0]$ and $p = 2$.

Main result: $\zeta = 56$ is the minimal period of the shift symmetric bit string generated by $Q = [4, 4, 2, 0]$ and $p = 2$, and $j = 18$ is the least even vector period of the corresponding vector representation.

Contraction results:

$$Q[2] = [4, 4, 2, 0]$$

$$\tau = [0, 3, 6, 7, 6], \quad r = [0, 1, 2, 3, 4]$$

$$\tau[r] = \tau[0, 1, 2, 3, 4] = [0, 3, 6, 7, 6]$$

$$Q[1] = [3 - 0, 6 - 3, 7 - 6, 6 - 7 + 1] = [3, 3, 1, 0]$$

$$\tau = [0, 2, 4, 4, 3], \quad r = [0, 1, 4]$$

$$\tau[r] = \tau[0, 1, 4] = [0, 2, 3]$$

$$Q[0] = [2 - 0, 3 - 2 + 1] = [2, 2]$$

Comment: $Q[2] = Q$ and $Q[i - 1]$ is the contraction vector of $Q[i]$ for $1 \leq i \leq 2$.

Periodic results:

$$p\text{-value} = 0, \quad Q[0] = [2, 2], \quad j[0] = 2, \quad \zeta[0] = 5$$

$$p\text{-value} = 1, \quad Q[1] = [3, 3, 1, 0], \quad j[1] = 18, \quad \zeta[1] = 38$$

$$p\text{-value} = 2, \quad Q[2] = [4, 4, 2, 0], \quad j[2] = 18, \quad \zeta[2] = 56$$

Comment: Suppose $0 \leq i \leq p$. Then $\zeta[i]$ is the minimal period of the shift symmetric bit string generated by $Q[i]$ and $p = i$, and $j[i]$ is the least even vector period of the corresponding vector representation.

Intermediate results:

Step 0: $j[0] = 2$ and $\zeta[0] = 5$ are the cyclic parameters of the extension $[2, 3]$ of $Q[0] = [2, 2]$.

Step 1:

$$Q[1] = [3, 3, 1, 0]$$

$$C[1] = [0, 3]$$

$$\tau = [0, 2, 4, 4, 3]$$

$$D[1] = [\tau(3)] = [4]$$

$$\gamma[1] = 1, \alpha[1] = 4, \gamma^*[1] = 1, \alpha^*[1] = 4$$

$$g = \gcd(\alpha^*[1], \zeta[0]) = \gcd(4, 5) = 1$$

$$x = \zeta[0]/g = 5/1 = 5, y = \alpha^*[1]/g = 4/1 = 4$$

$$j[1] = 2 \cdot x \cdot \gamma^*[1] + y \cdot j[0] = 2 \cdot 5 \cdot 1 + 4 \cdot 2 = 18$$

$$\zeta[1] = y \cdot \zeta[0] + j[1] = 4 \cdot 5 + 18 = 38$$

Step 2:

$$Q[2] = [4, 4, 2, 0], j[2] = j[1] = 18, \zeta[2] = \zeta[1] + j[2] = 38 + 18 = 56$$

Example 8.2.

Start values: $Q = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1]$ and $p = 1$.

Main result: $\zeta = 41$ is the minimal period of the shift symmetric bit string generated by $Q = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1]$ and $p = 1$, and $j = 26$ is the least even vector period of the corresponding vector representation.

Contraction results:

$$Q[1] = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1]$$

$$\tau = [0, 1, 1, 2, 2, 2, 4, 4, 5, 5, 5, 7, 7, 7, 8, 8, 8], r = [0, 5, 10, 13, 16]$$

$$\tau[r] = \tau[0, 5, 10, 13, 16] = [0, 2, 5, 7, 8]$$

$$Q[0] = [2 - 0, 5 - 2, 7 - 5, 8 - 7 + 1] = [2, 3, 2, 2]$$

Comment: $Q[1] = Q$ and $Q[0]$ is the contraction vector of $Q[1]$.

Periodic results:

$$p\text{-value} = 0, Q[0] = [2, 3, 2, 2], j[0] = 2, \zeta[0] = 5$$

$$p\text{-value} = 1, Q[1] = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1], j[1] = 26, \zeta[1] = 41$$

Comment: Suppose $0 \leq i \leq p$. Then $\zeta[i]$ is the minimal period of the shift symmetric bit string generated by $Q[i]$ and $p = i$, and $j[i]$ is the least even vector period of the corresponding vector representation.

Intermediate results:

Step 0: $j[0] = 2$ and $\zeta[0] = 5$ are the cyclic parameters of the extension $[2, 3, 2, 3]$ of $Q[0] = [2, 3, 2, 2]$.

Step 1:

$$\begin{aligned}
Q[1] &= [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1] \\
C[1] &= [0, 2, 4, 7, 9, 12, 15] \\
\tau &= [0, 1, 1, 2, 2, 2, 4, 4, 5, 5, 5, 7, 7, 7, 8, 8, 8] \\
D[1] &= [\tau(2), \tau(4), \tau(7), \tau(9), \tau(12), \tau(15)] = [1, 2, 4, 5, 7, 8] \\
\gamma[1] &= 6, \quad \alpha[1] = 9, \quad \gamma^*[1] = 2, \quad \alpha^*[1] = 3 \\
g &= \gcd(\alpha^*[1], \zeta[0]) = \gcd(3, 5) = 1 \\
x &= \zeta[0]/g = 5/1 = 5, \quad y = \alpha^*[1]/g = 3/1 = 3 \\
j[1] &= 2 \cdot x \cdot \gamma^*[1] + y \cdot j[0] = 2 \cdot 5 \cdot 2 + 3 \cdot 2 = 26 \\
\zeta[1] &= y \cdot \zeta[0] + j[1] = 3 \cdot 5 + 26 = 41
\end{aligned}$$

Example 8.3.

Start values: $Q = [3, 3, 3, 3, 2, 2, 1, 998]$ and $p = 2$.

Main result: $\zeta = 128653810$ is the minimal period of the shift symmetric bit string generated by $Q = [3, 3, 3, 3, 2, 2, 1, 998]$ and $p = 2$, and 1011778 is the least even vector period of the corresponding vector representation.

Contraction results:

$$\begin{aligned}
Q[2] &= [3, 3, 3, 3, 2, 2, 1, 998] \\
\tau &= [0, 2, 4, 6, 8, 9, 10, 10, 1007], \quad r = [0, 1, 2, 3, 4, 5, 8] \\
\tau[r] &= \tau[0, 1, 2, 3, 4, 5, 8] = [0, 2, 4, 6, 8, 9, 1007] \\
Q[1] &= (2 - 0, 4 - 2, 6 - 4, 8 - 6, 9 - 8, 1007 - 9 + 1) = [2, 2, 2, 2, 1, 999] \\
\tau &= [0, 1, 2, 3, 4, 4, 1002], \quad r = [0, 1, 2, 3, 6] \\
\tau[r] &= \tau[0, 1, 2, 3, 6] = [0, 1, 2, 3, 1002] \\
Q[0] &= (1 - 0, 2 - 1, 3 - 2, 1002 - 3 + 1) = [1, 1, 1, 1000] \\
\text{Comment: } Q[2] &= Q \text{ and } Q[i - 1] \text{ is the contraction vector of } Q[i] \text{ for } 1 \leq i \leq 2.
\end{aligned}$$

Periodic results:

$$\begin{aligned}
p\text{-value} = 0, \quad Q[0] &= [1, 1, 1, 1000], \quad j[0] = 4, \quad \zeta[0] = 1004 \\
p\text{-value} = 1, \quad Q[1] &= [2, 2, 2, 2, 1, 999], \quad j[1] = 6020, \quad \zeta[1] = 1013032 \\
p\text{-value} = 2, \quad Q[2] &= [3, 3, 3, 3, 2, 2, 1, 998], \quad j[2] = 1011778, \quad \zeta[2] = 128653810
\end{aligned}$$

Comment: Suppose $0 \leq i \leq p$. Then $\zeta[i]$ is the minimal period of the shift symmetric bit string generated by $Q[i]$ and $p = i$, and $j[i]$ is the least even vector period of the corresponding vector representation.

Intermediate results:

Step 0: $j[0] = 4$ and $\zeta[0] = 1004$ are the cyclic parameters of the extension $[1, 1, 1, 1001]$ of $Q[0] = [1, 1, 1, 1000]$.

Step 1:

$$Q[1] = [2, 2, 2, 2, 1, 999]$$

$$C[1] = [0, 5]$$

$$\tau = [0, 1, 2, 3, 4, 4, 1002]$$

$$D[1] = [\tau(5)] = [4]$$

$$\gamma[1] = 1, \alpha[1] = 1003, \gamma^*[1] = 1, \alpha^*[1] = 1003$$

$$g = \gcd(\alpha^*[1], \zeta[0]) = \gcd(1003, 1004) = 1$$

$$x = \zeta[0]/g = 1004/1 = 1004, \quad y = \alpha^*[1]/g = 1003/1 = 1003$$

$$j[1] = 2 \cdot x \cdot \gamma^*[1] + y \cdot j[0] = 2 \cdot 1004 \cdot 1 + 1003 \cdot 4 = 6020$$

$$\zeta[1] = y \cdot \zeta[0] + j[1] = 1003 \cdot 1004 + 6020 = 1013032$$

Step 2:

$$Q[2] = [3, 3, 3, 3, 2, 2, 1, 998]$$

$$C[2] = [0, 7]$$

$$\tau = [0, 2, 4, 6, 8, 9, 10, 10, 1007]$$

$$D[2] = [\tau(7)] = [10]$$

$$\gamma[2] = 1, \alpha[2] = 1008, \gamma^*[2] = 1, \alpha^*[2] = 1008$$

$$g = \gcd(\alpha^*[2], \zeta[1]) = \gcd(1008, 1013032) = 8$$

$$x = \zeta[1]/g = 1013032/8 = 126629, \quad y = \alpha^*[2]/g = 1008/8 = 126$$

$$j[2] = 2 \cdot x \cdot \gamma^*[2] + y \cdot j[1] = 2 \cdot 126629 \cdot 1 + 126 \cdot 6020 = 1011778$$

$$\zeta[2] = y \cdot \zeta[1] + j[2] = 126 \cdot 1013032 + 1011778 = 128653810.$$

9. Interpretation of the results.

We will interpret the periodic results in the examples in the previous section.

Example 9.1. In Example 8.1 the start values are $Q = [4, 4, 2, 0]$ and $p = 2$. By Example 8.1 we get the following periodic results:

$$p\text{-value} = 0, \quad Q[0] = [2, 2], \quad j[0] = 2, \quad \zeta[0] = 5$$

$$p\text{-value} = 1, \quad Q[1] = [3, 3, 1, 0], \quad j[1] = 18, \quad \zeta[1] = 38$$

$$p\text{-value} = 2, \quad Q[2] = [4, 4, 2, 0], \quad j[2] = 18, \quad \zeta[2] = 56$$

Let A_i^∞ be the shift symmetric bit string generated by $Q[i]$ and $p = i$ for $0 \leq i \leq 2$.

Let $A_0 = A(Q[0]) = A(2, 2) = 1_2 0_2 = 1100$,

$p_0 = 0$, $k_0 = w(A_0) - (p_0 + 1) = 2 - (0 + 1) = 1$ and $n_0 = \text{length}(A_0) = 4$.

Then A_0^∞ is generated from A_0 by the symmetric shift register with parameters k_0 , p_0 and n_0 . Moreover, $\zeta[0] = 5$ is the minimal period of A_0^∞ , and $j[0] = 2$ is the least even vector period of $V(A_0^\infty)$.

Let $A_1 = A(Q[1]) = A(3, 3, 1, 0) = 1_3 0_3 1_1 0_0 = 1110001$,

$p_1 = 1$, $k_1 = w(A_1) - (p_1 + 1) = 4 - (1 + 1) = 2$ and $n_1 = \text{length}(A_1) = 7$.

Then A_1^∞ is generated from A_1 by the symmetric shift register with parameters k_1 , p_1 and n_1 . Moreover, $\zeta[1] = 38$ is the minimal period of A_1^∞ , and $j[1] = 18$ is the least even vector period of $V(A_1^\infty)$.

Let $A_2 = A(Q[2]) = 1_4 0_4 1_2 0_0 = 1111000011$,

$p_2 = 2$, $k_2 = w(A_2) - (p_2 + 1) = 6 - (2 + 1) = 3$ and $n_2 = \text{length}(A_2) = 10$.

Then A_2^∞ is generated from A_2 by the symmetric shift register with parameters k_2 , p_2 and n_2 . Moreover, $\zeta[2] = 56$ is the minimal period of A_2^∞ , and $j[2] = 18$ is the least even vector period of $V(A_2^\infty)$.

Example 9.2. In Example 8.2 the start values are

$$Q = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1] \text{ and } p = 1.$$

By Example 8.2 we get the following periodic results:

$$p\text{-value} = 0, Q[0] = [2, 3, 2, 2], j[0] = 2, \zeta[0] = 5$$

$$p\text{-value} = 1, Q[1] = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1], j[1] = 26, \zeta[1] = 41$$

Suppose A_i^∞ is the shift-symmetric bit string generated by $Q[i]$ and $p = i$ for $i = 0$ and $i = 1$.

Let $A_0 = A(Q[0]) = 1_2 0_3 1_2 0_2 = 110001100$,

$p_0 = 0$, $k_0 = w(A_0) - (p_0 + 1) = 4 - (0 + 1) = 3$ and $n_0 = \text{length}(A_0) = 9$.

Then A_0^∞ is generated from A_0 by the symmetric shift register with parameters k_0 , p_0 and n_0 . Then $\zeta[0] = 5$ is the minimal period of A_0^∞ , and $j[0] = 2$ is the least even vector period of $V(A_0^\infty)$.

Let $A_1 = A(Q[1]) = 1_2 0_1 1_2 0_1 1_1 0_3 1_1 0_2 1_1 0_1 1_3 0_1 1_1 0_2 1_1 0_1$,

$p_1 = 1$, $k_1 = w(A) - (p_1 + 1) = 12 - (1 + 1) = 10$ and $n_1 = \text{length}(A) = 24$.

We note that $A_1 = 110110100010010111010010$. Moreover, A_1^∞ is generated from A_1 by the symmetric shift register with parameters k_1 , p_1 and n_1 . Then $\zeta[1] = 41$ is the minimal period of A_1^∞ , and $j[1] = 26$ is the least even vector period of $V(A_1^\infty)$.

Example 9.3. In Example 8.3 the start values are

$$Q = [3, 3, 3, 3, 2, 2, 1, 998] \text{ and } p = 2.$$

By Example 8.3 we get the following periodic results:

$$p\text{-value} = 0, Q[0] = [1, 1, 1, 1000], j[0] = 4, \zeta[0] = 1004$$

$$p\text{-value} = 1, Q[1] = [2, 2, 2, 2, 1, 999], j[1] = 6020, \zeta[1] = 1013032$$

$$p\text{-value} = 2, Q[2] = [3, 3, 3, 3, 2, 2, 1, 998], j[2] = 1011778, \zeta[2] = 128653810.$$

Suppose A_i^∞ is the shift-symmetric bit string generated by $Q[i]$ and $p = i$ for $0 \leq i \leq 2$.

$$\text{Let } A_0 = A(Q[0]) = 1_1 0_1 1_1 0_{1000} = 101 0_{1000},$$

$$p_0 = 0, k_0 = w(A_0) - (p_0 + 1) = 2 - (0 + 1) = 1 \text{ and } n_0 = \text{length}(A_0) = 1003.$$

Then A_0^∞ is generated from A_0 by the symmetric shift register with parameters k_0 , p_0 and n_0 . Moreover, $\zeta[0] = 1004$ is the minimal period of A_0^∞ , and $j[0] = 4$ is the least even vector period of $V(A_0^\infty)$.

$$\text{Let } A_1 = A(Q[1]) = 1_2 0_2 1_2 0_2 1_1 0_{999} = 110011001 0_{999},$$

$$p_1 = 1, k_1 = w(A_1) - (p_1 + 1) = 5 - (1 + 1) = 3 \text{ and } n_1 = \text{length}(A_1) = 1008.$$

Then A_1^∞ is generated from A_1 by the symmetric shift register with parameters k_1 , p_1 and n_1 . Moreover, $\zeta[1] = 1013032$ is the minimal period of A_1^∞ , and $j[1] = 6020$ is the least even vector period of $V(A_1^\infty)$.

$$\text{Let } A_2 = A(Q[2]) = 1_3 0_3 1_3 0_3 1_2 0_2 1_1 0_{998} = 11100011100011001 0_{998},$$

$$p_2 = 2, k_2 = w(A_2) - (p_2 + 1) = 9 - (2 + 1) = 6 \text{ and } n_2 = \text{length}(A_2) = 1015.$$

Then A_2^∞ is generated from A_2 by the symmetric shift register with parameters k_2 , p_2 and n_2 . Moreover, $\zeta[2] = 128653810$ is the minimal period of A_2^∞ , and $j[2] = 1011778$ is the least even vector period of $V(A_2^\infty)$.

Part 3.

10. The reduction to the main case.

In "Representation of symmetric shift registers" we describe how every case can be reduced to the main case. In fact, we prove that every case can be reduced to the situation where

$$(10.1) \quad A = a_1 \cdots a_n, \quad a_1 = 1, \quad w(A) = k + p + 1$$

and p is an admissible p -value of $Q = V(A)$.

Suppose (10.1) is true and A^∞ is generated from A by the symmetric shift register with parameters k , p and n . Let $Q = V(A)$. Then it is easily seen that $A = A(Q)$. Moreover, $k = w(A) - (p + 1)$. Hence, we get that A^∞ is

generated from $A = A(Q)$ by the symmetric shift register with parameters $k = w(A) - (p + 1)$, p and $n = \text{length}(A)$. That means, A^∞ is the shift symmetric bit string generated by Q and p .

By using Q and p as input, then the kotlin-program will calculate the minimal period ζ of A^∞ and the least even vector period j of the corresponding vector representation. Here is an example.

Example 10.1. Suppose A^∞ is generated from $A = 111000011000$ by the symmetric shift register with respect to $k = 2$, $p = 2$ and $n = 12$. We note that $w(A) = k + p + 1$ and A starts with 1. It is easily seen that $p = 2$ is an admissible p -value of $Q = V(A) = (3, 4, 2, 3)$. We refer to Observation 11.3. We note that $A = A(Q)$, $p = w(A) - (p + 1)$ and $n = \text{length}(A)$. Hence, A^∞ is the shift symmetric bit string generated by Q and p . By using $Q = (3, 4, 2, 3)$ and $p = 2$ as input, then the kotlin-program calculates the minimal period $\zeta = 94$ of A^∞ and the least even vector period $j = 26$ of the corresponding vector representation.

Part 4.

We define and give examples of admissible p -values.

11. Admissible p -values.

Suppose $p \geq 0$ and $V = (v_1, \dots, v_{J+1})$ is an even vector such that $v_i > 0$ for $1 \leq i \leq J$, $v_{J+1} \geq 0$ and J is odd. Let

$$(11.1) \quad \rho_0 = 0 \text{ and } \rho_j = v_1 - v_2 + v_3 - v_4 + \dots + (-1)^{j-1} v_j \text{ for } 1 \leq j \leq J + 1.$$

We call $\rho_0, \rho_1, \dots$ the alternating indexes of V . Then p is called an admissible p -value of V if there exists an odd integer t such that

$$1 \leq t \leq J, \rho_i > 0 \text{ for } 1 \leq i \leq t, \text{ and } \rho_t \geq p + 1.$$

Example 11.1. $p = 2$ is an admissible p -value of

$$V = (v_1, v_2, v_3, v_4) = (2, 1, 2, 0).$$

This is proved as follows:

$$\rho_1 = 2, \rho_2 = 2 - 1 = 1 \text{ and } \rho_3 = 2 - 1 + 2 = 3.$$

Let $t = 3$. Then $\rho_t = p + 1$ and $\rho_i > 0$ for $1 \leq i \leq t$.

Example 11.2. $p = 2$ is an admissible p -value of $V = (v_1, v_2) = (3, 1)$.

This is proved as follows: $\rho_1 = 3$. Let $t = 1$. Then $\rho_t = p + 1$ and $\rho_i > 0$ for $1 \leq i \leq t$.

Observation 11.3. Suppose $p \geq 0$. If V starts with a coordinate larger than p , then $\rho_1 \geq p + 1$ and p is an admissible p -value for V .

Proof. This is trivial (use $t = 1$).

12. Admissible p -values - an algorithmic approach.

Let $p \geq 0$ and $V = (v_1, \dots, v_{J+1})$ be as in the previous section. We note that $J = \#V - 1$ where $\#$ means the number of. Moreover, let $\rho_0, \rho_1, \dots$ be the alternating indexes of V .

Let i_{max} be maximal such that $1 \leq i_{max} \leq J$ and $\rho_i > 0$ for $1 \leq i \leq i_{max}$.

Let $\rho_{max} = \max\{\rho_i : 1 \leq i \leq i_{max}\}$. Then it can be proved that

(12.1) p is an admissible p -value of V if and only if $0 \leq p < \rho_{max}$.

Example 12.1. $V = (3, 2, 5, 2, 1, 7, 1, 1)$. Then $J = \#V - 1 = 8 - 1 = 7$. Moreover,

$$\begin{aligned}\rho_1 &= 3, \quad \rho_2 = 3 - 2 = 1, \quad \rho_3 = 3 - 2 + 5 = 6, \\ \rho_4 &= 3 - 2 + 5 - 2 = 4, \quad \rho_5 = 3 - 2 + 5 - 2 + 1 = 5, \\ \rho_6 &= 3 - 2 + 5 - 2 + 1 - 7 = -2.\end{aligned}$$

We note that $i_{max} = 5$ and $\rho_{max} = \rho_3 = 6$. We can therefore conclude that $p = 0, p = 1, \dots, p = 5$ are the admissible p -values of V .

Example 12.2. $V = (3, 4, 2, 4, 1, 0)$. In this case $J = \#V - 1 = 6 - 1 = 5$. Moreover, $\rho_1 = 3$ and $\rho_2 = 3 - 4 = -1$. We note that $i_{max} = 1$ and $\rho_{max} = \rho_1 = 3$. Hence, we get that $p = 0, p = 1$ and $p = 2$ are the admissible p -values of V .

Example 12.3. $V = (2, 3, 1, 3)$. In this case $J = \#V - 1 = 4 - 1 = 3$. Moreover, $\rho_1 = 2$ and $\rho_2 = 2 - 3 = -1$. We note that $i_{max} = 1$ and $\rho_{max} = \rho_1 = 2$. Hence, we get that $p = 0$ and $p = 1$ are the admissible p -values of V .

Example 12.4. $V = (1, 5)$. In this case $J = \#V - 1 = 2 - 1 = 1$. Moreover, $\rho_1 = 1$. Hence $i_{max} = 1$ and $\rho_{max} = \rho_1 = 1$. Hence, we get that $p = 0$ is the only admissible p -value of V .

Part 5.

We define the distance measure, the distance function τ and the τ -vector.

13. The distance measure.

If v is an integer, let $v^- = v - 1$. If V is a vector, let $\delta(V) = \text{sum}(V) - \#V$. In particular, if $V = \emptyset$, then $\delta(V) = 0$. We note that

$$\delta(V) = \text{sum}(V) - s = v_1^- + \dots + v_s^- \text{ if } V = (v_1, \dots, v_s) \text{ where } s \geq 1.$$

Suppose $V = (v_1, \dots, v_s)$ where $v_j \geq 1$ for $1 \leq j \leq s$. Then $\delta(V) \geq 0$. If in addition one of the coordinates of V are larger than 1, then $\delta(V) > 0$.

Example 13.1. If $G_1 = (5)$, then $\delta(G_1) = 5 - 1 = 4$.

If $G_2 = (3, 1, 2, 1, 5, 1, 4)$, then $\delta(G_2) = \text{sum}(G_2) - 7 = 17 - 7 = 10$.

If $G_3 = (4, 1, 3, 1, 1)$, then $\delta(G_3) = \text{sum}(G_3) - 5 = 10 - 5 = 5$.

14. The distance function.

We let τ correspond to *tau* in the kotlin-program in order to get a smoother presentation.

Suppose $Q = (v_1, \dots, v_s)$. The distance function τ of Q is given by

$$(14.1) \quad \tau(0) = 0 \text{ and } \tau(r) = \delta(v_1, \dots, v_r) = v_1^- + \dots + v_r^- \text{ for } 1 \leq r \leq s.$$

We note that $\tau(r+1) = \tau(r) + v_{r+1}^-$ for $0 \leq r < s$, and $\delta(Q) = \tau(s)$,

$$(14.2) \quad \tau(t) = \tau(r) + \delta(v_{r+1}, \dots, v_t) \text{ if } 0 \leq r < t \leq s.$$

The vector $\tau = (\tau(0), \dots, \tau(s))$ is called the τ -vector of Q . The coordinates are the function values of τ . We use τ to denote both the function and the vector. However, it is always clear from the context whether τ is the function or the vector. The dependence of Q will also be clear from the context.

15. Examples.

If $Q = (v_1, \dots, v_s)$, then $\tau(r+1) = \tau(r) + v_{r+1}^-$ for $0 \leq r < s$. In the next examples with the exception of Example 15.2 we use this formula.

Example 15.1. Let $Q = (v_1, \dots, v_4) = (4, 4, 2, 0)$. Then $\tau = (0, 3, 6, 7, 6)$ since $\tau(0) = 0$,

$$\tau(1) = \tau(0) + v_1^- = 0 + 4^- = 3$$

$$\tau(2) = \tau(1) + v_2^- = 3 + 4^- = 6$$

$$\tau(3) = \tau(2) + v_3^- = 6 + 2^- = 7$$

$$\tau(4) = \tau(3) + v_4^- = 7 + 0^- = 6.$$

Example 15.2. Suppose $Q = (3, 3, 1, 0)$. Then $\tau = (0, 2, 4, 4, 3)$ since

$$\tau(0) = 0$$

$$\tau(1) = \delta(3) = 3^- = 2$$

$$\tau(2) = \delta(3, 3) = 3^- + 3^- = 2 + 2 = 4$$

$$\tau(3) = \delta(3, 3, 1) = 3^- + 3^- + 1^- = 2 + 2 + 0 = 4$$

$$\tau(4) = \delta(3, 3, 1, 0) = 3^- + 3^- + 1^- + 0^- = 2 + 2 + 0 - 1 = 3.$$

Example 15.3. $Q = (v_1, \dots, v_{16}) = (2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1)$. Then $\tau = (0, 1, 1, 2, 2, 2, 4, 4, 5, 5, 5, 7, 7, 7, 8, 8)$ since

$$\tau(0) = 0, \quad \tau(1) = \tau(0) + v_1^- = 0 + 2^- = 1$$

$$\tau(2) = \tau(1) + v_2^- = 1 + 1^- = 1, \quad \tau(3) = \tau(2) + v_3^- = 1 + 2^- = 2$$

$$\tau(4) = \tau(3) + v_4^- = 2 + 1^- = 2, \text{ etc.}$$

Example 15.4. Suppose $Q = (v_1, v_2, \dots, v_8) = (3, 3, 3, 3, 2, 2, 1, 998)$. Then we get that $\tau = (0, 2, 4, 6, 8, 9, 10, 10, 1007)$ since

$$\begin{aligned}\tau(0) &= 0, \\ \tau(1) &= \tau(0) + v_1^- = 0 + 3^- = 2 \\ \tau(2) &= \tau(1) + v_2^- = 2 + 3^- = 4 \\ \tau(3) &= \tau(2) + v_3^- = 4 + 3^- = 6 \\ \tau(4) &= \tau(3) + v_4^- = 6 + 3^- = 8 \\ \tau(5) &= \tau(4) + v_5^- = 8 + 2^- = 9, \text{ etc.}\end{aligned}$$

Example 15.5. Suppose $Q = (v_1, v_2, \dots, v_6) = (2, 2, 2, 2, 1, 999)$. Then we get that $\tau = (0, 1, 2, 3, 4, 4, 1002)$ since

$$\begin{aligned}\tau(0) &= 0, \\ \tau(1) &= \tau(0) + v_1^- = 0 + 2^- = 1 \\ \tau(2) &= \tau(1) + v_2^- = 1 + 2^- = 2 \\ \tau(3) &= \tau(2) + v_3^- = 2 + 2^- = 3 \\ \tau(4) &= \tau(3) + v_4^- = 3 + 2^- = 4 \\ \tau(5) &= \tau(4) + v_5^- = 4 + 1^- = 4 \\ \tau(6) &= \tau(5) + v_6^- = 4 + 999^- = 1002.\end{aligned}$$

Part 6.

We will define the r -indexes, r -vector and the contraction vector π . In this part we suppose $Q = (v_1, \dots, v_{s-1}, v_s)$ where

$$s \text{ is even, } v_1 > 1, v_i \geq 1 \text{ for } 2 \leq i \leq s-1, \text{ and } v_s \geq 0.$$

We note that $s = \#Q$ where $\#Q$ denotes "the number of."

16. Auxiliary functions.

Suppose $0 \leq r \leq s-1$. Let $t_{\max}(r) = t$ and $\text{next}(r) = r + 2t + 1$ where $t \geq 0$ is maximal such that

$$v_{r+2i} = 1 \text{ for } 1 \leq i \leq t, \text{ and } r + 2t \leq s-1.$$

We note that

$$\text{next}(r) = r + 2t_{\max}(r) + 1 \text{ for } 0 \leq r \leq s-1.$$

In particular, $\text{next}(r) = r + 1$ if $t_{\max}(r) = 0$. It is easily seen that the next observation is true.

- Observation 16.1.** a) If $r = s-1$, then $t_{\max}(r) = 0$ and $\text{next}(r) = r+1 = s$.
b) If $r = s-2$, then $t_{\max}(r) = 0$ and $\text{next}(r) = r+1 = s-1$.
c) If $v_{r+2} > 1$ where $0 \leq r < s-2$, then $t_{\max}(r) = 0$ and $\text{next}(r) = r+1$.

17. The r -indexes.

We choose $r_0 = 0 < r_1 < \cdots < r_{I+1} = s$ such that $r_{j+1} = \text{next}(r_j)$ for $0 \leq j \leq I$. Then $r_0, \dots, r_{I+1}$ are the r -indexes and $r = (r_0, \dots, r_{I+1})$ the r -vector of Q .

Example 17.1. Suppose $Q = (v_1, \dots, v_4) = (4, 4, 2, 0)$. We note in this case that $s = \#Q = 4$ and $s - 1 = 3$. Suppose $0 \leq r \leq 3$. Then $t_{\max}(r) = t$ where $t \geq 0$ is maximal such that $v_{r+2i} = 1$ for $1 \leq i \leq t$, and $r + 2t \leq 3$.

Let $r_0 = 0$. By Observation 16.1 c) we get that

$$r_1 = \text{next}(r_0) = \text{next}(0) = 1 \text{ since } v_{0+2} = v_2 = 4 > 1$$

$$r_2 = \text{next}(r_1) = \text{next}(1) = 2 \text{ since } v_{1+2} = v_3 = 2 > 1$$

Moreover, by Observation 16.1 b) and a) we get that

$$r_3 = \text{next}(r_2) = \text{next}(2) = 3 \text{ since } 2 = s - 2$$

$$r_4 = \text{next}(r_3) = \text{next}(3) = 4 \text{ since } 3 = s - 1$$

Hence, $r = (r_0, \dots, r_4) = (0, 1, 2, 3, 4)$.

Example 17.2. Suppose $Q = (v_1, \dots, v_4) = (3, 3, 1, 0)$. In this case we note that $s = \#Q = 4$ and $s - 1 = 3$. If $0 \leq r \leq 3$, then $t_{\max}(r) = t$ where $t \geq 0$ is maximal such that $v_{r+2i} = 1$ for $1 \leq i \leq t$, and $r + 2t \leq 3$.

Let $r_0 = 0$. By Observation 16.1 c) we get that

$$r_1 = \text{next}(r_0) = \text{next}(0) = 1 \text{ since } v_{0+2} = v_2 = 3 > 1$$

Moreover, $t_{\max}(r_1) = 1$ since $v_{r_1+2} = v_3 = 1$ and $r_1 + 2 \cdot t > 3$ for $t > 1$. Hence, $r_2 = r_1 + 2t_{\max}(r_1) + 1 = 1 + 2 \cdot 1 + 1 = 4$, and $r = (r_0, r_1, r_2) = (0, 1, 4)$.

Example 17.3. Suppose

$$Q = (v_1, \dots, v_{16}) = (2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1).$$

We note that $s = \#Q = 16$ and $s - 1 = 15$. If $0 \leq r \leq 15$, then $t_{\max}(r) = t$ where $t \geq 0$ is maximal such that $v_{r+2i} = 1$ for $1 \leq i \leq t$, and $r + 2t \leq 15$.

Let $r_0 = 0$. Next, we get that

$$t_{\max}(r_0) = 2 \text{ since } v_{r_0+2} = v_2 = 1, v_{r_0+4} = v_4 = 1 \text{ and } v_{r_0+6} = v_6 = 3 > 1.$$

$$r_1 = r_0 + 2t_{\max}(r_0) + 1 = 0 + 2 \cdot 2 + 1 = 5$$

$$t_{\max}(r_1) = 2 \text{ since } v_{r_1+2} = v_7 = 1, v_{r_1+4} = v_9 = 1 \text{ and } v_{r_1+6} = v_{11} = 3 > 1.$$

$$r_2 = r_1 + 2t_{\max}(r_1) + 1 = 5 + 2 \cdot 2 + 1 = 10$$

$$t_{\max}(r_2) = 1 \text{ since } v_{r_2+2} = v_{12} = 1 \text{ and } v_{r_2+4} = v_{14} = 2 > 1$$

$$r_3 = r_2 + 2t_{\max}(r_2) + 1 = 10 + 2 \cdot 1 + 1 = 13$$

$t_{max}(r_3) = 1$ since $v_{r_3+2} = v_{15} = 1$ and $v_{r_3+4} > 15$

$$r_4 = r_3 + 2t_{max}(r_3) + 1 = 13 + 2 \cdot 1 + 1 = 16$$

Hence, we can conclude that $r = (r_0, \dots, r_4) = (0, 5, 10, 13, 16)$.

Example 17.4. Suppose $Q = (v_1, \dots, v_6) = (2, 2, 2, 2, 1, 999)$. Let $r_0 = 0$.

Since $v_{i+2} > 1$ for $0 \leq i \leq 2$, then Observation 16.1 c) implies that

$$r_{i+1} = next(r_i) = next(i) = i + 1 \text{ for } 0 \leq i \leq 2.$$

In particular, we note that $r_3 = 3$. Since $v_5 = 1$, then we get that $t_{max}(3) = 1$ and $r_4 = next(r_3) = next(3) = 3 + 2 \cdot 1 + 1 = 6$. We can therefore conclude that $r = (r_0, \dots, r_4) = (0, 1, 2, 3, 6)$.

Example 17.5. $Q = (v_1, \dots, v_8) = (3, 3, 3, 3, 2, 2, 1, 998)$. Let $r_0 = 0$.

Since $v_{i+2} > 1$ for $0 \leq i \leq 4$, then Observation 16.1 c) implies that

$$r_{i+1} = next(r_i) = next(i) = i + 1 \text{ for } 0 \leq i \leq 4.$$

In particular, $r_5 = 5$. Since $v_7 = 1$, then

$$t_{max}(5) = 1 \text{ and } r_6 = next(r_5) = next(5) = 5 + 2 \cdot 1 + 1 = 8.$$

Hence, $r = (r_0, \dots, r_6) = (0, 1, 2, 3, 4, 5, 8)$.

18. The contraction vector.

Suppose $r_0 = 0 < r_1 < \dots < r_{I+1} = s$ are the r -indexes of $Q = (v_1, \dots, v_s)$, and $r = (r_0, \dots, r_{I+1})$. Let τ be the distance function of Q . Moreover, let

$$\tau(r) = (\tau(r_0), \tau(r_1), \dots, \tau(r_I), \tau(r_{I+1})).$$

Then the contraction vector $\pi(Q)$ of Q is given by

$$\pi(Q) = (\tau(r_1) - \tau(r_0), \tau(r_2) - \tau(r_1), \dots, \tau(r_I) - \tau(r_{I-1}), \tau(r_{I+1}) - \tau(r_I) + 1).$$

Example 18.1. $Q = (v_1, \dots, v_4) = (4, 4, 2, 0)$. By Example 15.1 and 17.1 we get that

$$\tau = (0, 3, 6, 7, 6) \text{ and } r = (r_0, \dots, r_4) = (0, 1, 2, 3, 4).$$

Hence,

$$\begin{aligned} \tau(r) &= (\tau(0), \tau(1), \tau(2), \tau(3), \tau(4)) = (0, 3, 6, 7, 6), \\ \pi(Q) &= (\tau(r_1) - \tau(r_0), \tau(r_2) - \tau(r_1), \tau(r_3) - \tau(r_2), \tau(r_4) - \tau(r_3) + 1) \\ &= (3 - 0, 6 - 3, 7 - 6, 6 - 7 + 1) = (3, 3, 1, 0). \end{aligned}$$

Example 18.2. Suppose $Q = (v_1, \dots, v_4) = (3, 3, 1, 0)$. By Example 15.2 and 17.2 we get that $\tau = (0, 2, 4, 4, 3)$ and $r = (r_0, r_1, r_2) = (0, 1, 4)$. Hence,

$$\begin{aligned} \tau(r) &= (\tau(0), \tau(1), \tau(4)) = (0, 2, 3), \\ \pi(Q) &= (\tau(r_1) - \tau(r_0), \tau(r_2) - \tau(r_1) + 1) = (2 - 0, 3 - 2 + 1) = (2, 2). \end{aligned}$$

Example 18.3. Let $Q = (v_1, \dots, v_{16}) = (2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1)$. By Example 15.3 and 17.3 we get $\tau = (0, 1, 1, 2, 2, 2, 4, 4, 5, 5, 5, 7, 7, 7, 8, 8, 8)$ and $r = (r_0, \dots, r_4) = (0, 5, 10, 13, 16)$. Hence,

$$\begin{aligned}\tau(r) &= (\tau(0), \tau(5), \tau(10), \tau(13), \tau(16)) = (0, 2, 5, 7, 8), \\ \pi(Q) &= (\tau(r_1) - \tau(r_0), \tau(r_2) - \tau(r_1), \tau(r_3) - \tau(r_2), \tau(r_4) - \tau(r_3) + 1) \\ &= (2 - 0, 5 - 2, 7 - 5, 8 - 7 + 1) = (2, 3, 2, 2).\end{aligned}$$

Example 18.4. Suppose $Q = (v_1, \dots, v_8) = (3, 3, 3, 3, 2, 2, 1, 998)$. By Example 15.4 and 17.4 we get that

$$\begin{aligned}\tau &= (0, 2, 4, 6, 8, 9, 10, 10, 1007) \text{ and } r = (r_0, \dots, r_6) = (0, 1, 2, 3, 4, 5, 8), \\ \tau(r) &= (\tau(0), \tau(1), \dots, \tau(5), \tau(8)) = (0, 2, 4, 6, 8, 9, 1007), \\ \pi(Q) &= (\tau(r_1) - \tau(r_0), \dots, \tau(r_5) - \tau(r_4), \tau(r_6) - \tau(r_5) + 1) \\ &= (2 - 0, 4 - 2, 6 - 4, 8 - 6, 9 - 8, 1007 - 9 + 1) = (2, 2, 2, 2, 1, 999).\end{aligned}$$

Example 18.5. Suppose $Q = (v_1, \dots, v_6) = (2, 2, 2, 2, 1, 999)$. By using Example 15.5 and 17.5 we get that

$$\begin{aligned}\tau &= (0, 1, 2, 3, 4, 4, 1002) \text{ and } r = (r_0, \dots, r_4) = (0, 1, 2, 3, 6), \\ \tau(r) &= (\tau(0), \tau(1), \tau(2), \tau(3), \tau(6)) = (0, 1, 2, 3, 1002), \\ \pi(Q) &= (\tau(r_1) - \tau(r_0), \tau(r_2) - \tau(r_1), \tau(r_3) - \tau(r_2), \tau(r_4) - \tau(r_3) + 1) \\ &= (1 - 0, 2 - 1, 3 - 2, 1002 - 3 + 1) = (1, 1, 1, 1000).\end{aligned}$$

Part 7.

We will define the extension of vectors and cyclic parameters.

If $V = (v_1, v_2, \dots, v_s)$, then $\text{sum}(V, j) = v_1 + \dots + v_j$ for $1 \leq j \leq s$.

19. The extension of vectors.

If Q is a vector, then the extension Q^* of Q is constructed by adding one to the last coordinate of Q . For example, the extension of $Q = (1, 1, 3, 2, 2, 4)$ is given by $Q^* = (1, 1, 3, 2, 2, 4 + 1) = (1, 1, 3, 2, 2, 5)$.

20. Cyclic parameters.

If $V = (v_1, \dots, v_r)$ where $r > 0$, let $\psi(V) = (v_2, \dots, v_r, v_1)$. It is easily seen that $\psi^r(V) = V$ and $\psi^j(V) = (v_{j+1}, \dots, v_r, v_1, \dots, v_j)$ for $1 \leq j < r$.

We suppose $V = (v_1, \dots, v_r)$ where $r > 0$ is even. Since $\psi^r(V) = V$, there exists a least even positive integer $j \leq r$ such that $\psi^j(V) = V$.

Let $\zeta = \text{sum}(V, j) = v_1 + \dots + v_j$. We call j and ζ the cyclic parameters of V . It is easily proved that j is an even positive factor of $r = \#V$.

Example 20.1. Let $V = (2, 1, 1, 4, 2, 1, 1, 4)$. Since j must be an even positive factor of $\#V = 8$, there are three possibilities, $j = 2$, $j = 4$ or $j = 8$. We get that $\psi^2(V) = (1, 4, 2, 1, 1, 4, 2, 1) \neq V$ and $\psi^4(V) = V$.

Then $j = 4$ is the least even positive integer such that $\psi^j(V) = V$. Hence, $j = 4$ and $\zeta = \text{sum}(V, 4) = 2 + 1 + 1 + 4 = 8$ are the cyclic parameters of V .

Example 20.2. Suppose $V = (1, 1, 1, 1001)$. Since j must be an even positive factor of $\#V = 4$, there are two possibilities, $j = 2$ or $j = 4$. We note that $\psi^2(V) = (1, 1001, 1, 1) \neq V$. Hence, $j = 4$ is the least even positive integer such that $\psi^j(V) = V$. Then $j = 4$ and $\zeta = \text{sum}(V, 4) = 1004$ are the cyclic parameters of V .

Example 20.3. Suppose $V = (2, 3)$. Since $\#V = 2$, then $j = 2$ is the least even positive integer such that $\psi^j(V) = V$. Hence,

$j = 2$ and $\zeta = \text{sum}(V, 2) = 2 + 3 = 5$ are the cyclic parameters of V .

Example 20.4. Suppose $V = (2, 3, 2, 3)$. Since j must be an even positive factor of $\#V = 4$, there are two possibilities, $j = 2$ or $j = 4$.

We get that $\psi^2(V) = (2, 3, 2, 3) = V$.

Then $j = 2$ is the least even positive integer such that $\psi^j(V) = V$. Hence, $j = 2$ and $\zeta = \text{sum}(V, 2) = 2 + 3 = 5$ are the cyclic parameters of V .

Part 8.

We will define the parameter α and the distance vector.

21. The parameter α .

Suppose $Q = (v_1, \dots, v_s)$. Then $\alpha = \delta(Q) + 1 = v_1^- + \dots + v_s^- + 1$. The parameter depends on Q but this will always be clear from the context. We note that $\alpha = \tau(s) + 1$ where τ is the distance function of Q . We can also calculate α as follows:

$$\alpha = \delta(Q) + 1 = \text{sum}(Q) - s + 1 = \text{sum}(Q) - (s - 1) \text{ where } s = \#Q.$$

Example 21.1. Suppose $Q = (3, 3, 1, 0)$. Then

$$\alpha = (3^- + 3^- + 1^- + 0^-) + 1 = (2 + 2 + 0 - 1) + 1 = 4.$$

Example 21.2. Suppose $Q = (2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1)$. We note that $s = \#Q = 16$ and $s - 1 = 15$. Hence,

$$\alpha = (2 + 1 + 2 + 1 + 1 + 3 + 1 + 2 + 1 + 1 + 3 + 1 + 1 + 2 + 1 + 1) - 15 = 9.$$

22. The distance vector.

Suppose $Q = (v_1, \dots, v_s)$ where

$$s > 1 \text{ is even, } v_1 > 1, v_i \geq 1 \text{ for } 2 \leq i \leq s-1, \text{ and } v_s \geq 0.$$

We also suppose there exists a coordinate $v_i = 1$ where $2 \leq i \leq s-1$. Hence, we can in this case choose indexes $c_0, \dots, c_\gamma$ such that

$$(22.1) \quad c_0 = 0 < c_1 < \dots < c_\gamma < s,$$

$$(22.2) \quad c_{i+1} \text{ is the least index } > c_i + 1 \text{ such that } v_{c_{i+1}} = 1 \text{ for } 0 \leq i < \gamma,$$

$$(22.3) \quad v_c > 1 \text{ for } c_\gamma + 1 < c < s.$$

Since $c_0 = 0$, then c_1 is the least index > 1 such that $v_{c_1} = 1$.

We call $c_0, \dots, c_\gamma$ the c -indexes of V . Let $C = (c_0, \dots, c_\gamma)$ be the vector containing the c -indexes. We call C the index vector of Q .

We define the distance vector of Q as follows:

$$(22.4) \quad D = (\tau(c_1), \dots, \tau(c_\gamma)) = (d_1, \dots, d_\gamma)$$

where $d_i = \tau(c_i)$ for $1 \leq i \leq \gamma$. We note that $\#D = \gamma$. Alternatively, γ is the number of positive c -indexes. Let $\alpha = \delta(Q) + 1$.

Since $v_1 > 1$, then it can be proved that

$$(22.5) \quad 0 < d_1 \leq d_2 \leq \dots \leq d_\gamma \leq \alpha \text{ where } \alpha = \delta(Q) + 1.$$

In the following examples the coordinates corresponding to the positive c -indexes are overlined. We also calculate α in each example.

Example 22.1. Suppose $Q = (3, 3, \bar{1}, 0)$. Then $c_0 = 0$ and $c_1 = 3$ are the c -index of Q . In this case $\gamma = 1$, $C = (0, 3)$ and $D = (\tau(3)) = (4)$. Moreover, we get that $\alpha = \delta(Q) + 1 = 4$.

Example 22.2. Suppose $Q = (2, \bar{1}, 2, \bar{1}, 1, 3, \bar{1}, 2, \bar{1}, 1, 3, \bar{1}, 1, 2, \bar{1}, 1)$. Then

$$c_0 = 0, c_1 = 2, c_2 = 4, c_3 = 7, c_4 = 9, c_5 = 12 \text{ and } c_6 = 15$$

are the c -index of Q . In this case $\gamma = 6$, $\alpha = \delta(Q) + 1 = 9$,

$$C = (0, 2, 4, 7, 9, 12, 15),$$

$$\tau = (0, 1, 1, 2, 2, 2, 4, 4, 5, 5, 5, 7, 7, 7, 8, 8, 8),$$

$$D = (\tau(2), \tau(4), \tau(7), \tau(9), \tau(12), \tau(15)) = (1, 2, 4, 5, 7, 8)$$

Example 22.3. Suppose $Q = (3, 3, 3, 3, 2, 2, \bar{1}, 998)$. Then $c_0 = 0$ and $c_1 = 7$ are the c -indexes of Q . Hence, $C = (0, 7)$ and $D = (\tau(c_1)) = (10)$ since $\tau(c_1) = \tau(7) = \delta(3, 3, 3, 3, 2, 2, 1) = 10$. Moreover, $\alpha = \delta(Q) + 1 = 1008$.

Example 22.4. Suppose $Q = (2, 2, 2, 2, \bar{1}, 999)$. Then $c_0 = 0$ and $c_1 = 5$ are the c -index of Q . Hence, $C = (0, 5)$ and $D = (\tau(c_1)) = (4)$ since $\tau(c_1) = \tau(5) = \delta(2, 2, 2, 2, 1) = 4$. Moreover, $\alpha = \delta(Q) + 1 = 1003$.

Part 9.

23. Progression parameters.

In this section we suppose $\alpha > 0$ and

$$D = (d_1, d_2, \dots, d_\gamma) \text{ where } 0 < d_1 \leq d_2 \leq \dots \leq d_\gamma \leq \alpha.$$

We note that $\gamma = \#D$ = the number of coordinates of D . Let $E = (D, D + \alpha)$. That means, $E = (e_1, \dots, e_{2\gamma})$ where $e_i = d_i$ and $e_{\gamma+i} = d_i + \alpha$ for $1 \leq i \leq \gamma$.

Let $F = \{m > 0 : m \text{ factor of } \gcd(\alpha, \gamma)\}$.

Suppose $m \in F$. Let $\beta = \frac{\alpha}{m}$ and $r = \frac{\gamma}{m}$. If $e_{r+i} = d_i + \beta$ for $1 \leq i \leq \gamma$, then m is called a progression coefficient of D with respect to α .

Alternatively, m is called a progression coefficient of D with respect to α if $(e_{r+1}, \dots, e_{r+\gamma}) = (d_1, \dots, d_\gamma) + \beta$.

By Observation 23.1 we get that $m = 1$ is a progression coefficient of D with respect to α . Let m^* be the maximal progression coefficient of D with respect to α . Then $\alpha^* = \frac{\alpha}{m^*}$ and $\gamma^* = \frac{\gamma}{m^*}$ are called the least progression parameters of D with respect to α .

Let $m_1 > m_2 > \dots > m_j = 1$ be the factors in F in descending order. Then we test successively if m_i is a progression coefficient of D with respect to α for $i = 1, i = 2, \dots$, until we find i such that

$$(e_{r+1}, \dots, e_{r+\gamma}) = (d_1, \dots, d_\gamma) + \beta \text{ where } \beta = \frac{\alpha}{m_i} \text{ and } r = \frac{\gamma}{m_i}.$$

Then $m^* = m_i$ is the maximal progression coefficient of D with respect to α , and $\alpha^* = \frac{\alpha}{m^*} = \beta$ and $\gamma^* = \frac{\gamma}{m^*} = r$.

Observation 23.1. $m = 1$ is a progression coefficient of D with respect to α .

Proof. If $m = 1$, then $\beta = \frac{\alpha}{m} = \alpha$ and $r = \frac{\gamma}{m} = \gamma$. It is sufficient to prove that $e_{r+i} = e_i + \beta$ for $1 \leq i \leq \gamma$. This is true since

$$e_{r+i} = e_{\gamma+i} = d_i + \alpha = e_i + \beta \text{ for } 1 \leq i \leq \gamma.$$

Observation 23.2. Suppose $\gcd(\alpha, \gamma) = 1$.

a) $m^* = 1$ is the maximal progression coefficient of D with respect to α .

b) $\alpha^* = \frac{\alpha}{1} = \alpha$ and $\gamma^* = \frac{\gamma}{1} = \gamma$ are the least progression parameters of D with respect to α .

Proof. a) If $\gcd(\alpha, \gamma) = 1$, then $F = \{m > 0 : m \text{ factor of } \gcd(\alpha, \gamma)\} = \{1\}$. By Observation 16.1 we get that $m^* = 1$ is the only progression coefficient of D with respect to α .

b) follows from a).

In particular, if $\gamma = 1$, then $\gcd(\alpha, \gamma) = 1$ and Observation 23.2 implies that $\alpha^* = \frac{\alpha}{1} = \alpha$ and $\gamma^* = \frac{\gamma}{1} = \gamma = 1$.

Example 23.3. Let $D = (d_1, \dots, d_6) = (2, 4, 8, 10, 14, 16)$ and $\alpha = 18$. Then $\gamma = \#D = 6$ and $\gcd(\alpha, \gamma) = 6$. In this case $F = \{6, 3, 1\}$ and

$$E = (D, D + \alpha) = (e_1, \dots, e_{12}) = (2, 4, 8, 10, 14, 16, 20, 22, 26, 28, 32, 34).$$

Suppose $m = 6$. Let $\beta = \frac{\alpha}{m} = 3$ and $r = \frac{\gamma}{m} = 1$. Then

$$(e_{r+1}, \dots, e_{r+6}) = (e_2, \dots, e_7) = (4, 8, 10, 14, 16, 20) \neq (d_1, \dots, d_6) + \beta.$$

Suppose $m = 3$. Let $\beta = \frac{\alpha}{m} = 6$ and $r = \frac{\gamma}{m} = 2$. Then

$$(e_{r+1}, \dots, e_{r+6}) = (e_3, \dots, e_8) = (8, 10, 14, 16, 20, 22) = (d_1, \dots, d_6) + \beta.$$

Hence, $m^* = 3$ is the maximal progression coefficient of D with respect to α , and $\alpha^* = \frac{\alpha}{m^*} = 6$ and $\gamma^* = \frac{\gamma}{m^*} = 2$.

Example 23.4. Let $D = (d_1, d_2, d_3) = (1, 6, 11)$ and $\alpha = 15$. Then we get $\gamma = \#D = 3$ and $E = (D, D + \alpha) = (e_1, \dots, e_6) = (1, 6, 11, 16, 21, 26)$. The positive factors of $\gcd(\alpha, \gamma) = 3$ are 3 and 1. In this case $F = \{3, 1\}$.

Suppose $m = 3$. Let $\beta = \frac{\alpha}{m} = \frac{15}{3} = 5$ and $r = \frac{\gamma}{m} = \frac{3}{1} = 1$. Then

$$(e_{r+1}, e_{r+2}, e_{r+3}) = (e_2, e_3, e_4) = (6, 11, 16) = (d_1, d_2, d_3) + \beta.$$

We conclude that $m^* = 3$ is the maximal progression coefficient of D with respect to α , and $\alpha^* = \frac{\alpha}{m^*} = 5$ and $\gamma^* = \frac{\gamma}{m^*} = 1$.

Example 23.5. Let $D = (d_1, \dots, d_6) = (1, 2, 4, 5, 7, 8)$ and $\alpha = 9$. Then $\gamma = \#D = 6$ and $\gcd(\alpha, \gamma) = 3$. In this case $F = \{3, 1\}$ and

$$E = (e_1, \dots, e_{12}) = (1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17).$$

Suppose $m = 3$. Let $\beta = \frac{\alpha}{m} = 3$ and $r = \frac{\gamma}{m} = 2$. Then

$$(e_{r+1}, \dots, e_{r+6}) = (e_3, \dots, e_8) = (4, 5, 7, 8, 10, 11) = (d_1, \dots, d_6) + \beta.$$

Hence, $m^* = 3$ is the maximal progression coefficient of D with respect to α , and $\alpha^* = \frac{\alpha}{m^*} = 3$ and $\gamma^* = \frac{\gamma}{m^*} = 2$.

Part 10.

We will describe how the output results in Example 8.1, 8.2 and 8.3 are deduced. The calculation of the parameters and vectors are contained in the examples in the previous sections. But it is necessary to explain where the relevant calculations are carried out. This will be done in this part.

Example 8.1 - continued.

Start values: $Q = [4, 4, 2, 0]$ and $p = 2$. Since the first coordinate is larger than p , then Observation 11.3 implies that p is an admissible p -value of Q .

Contraction results:

$$\begin{aligned}
Q[2] &= [4, 4, 2, 0] \\
\tau &= [0, 3, 6, 7, 6], \quad r = [0, 1, 2, 3, 4] \\
\tau[r] &= \tau[0, 1, 2, 3, 4] = [0, 3, 6, 7, 6] \\
Q[1] &= [3 - 0, 6 - 3, 7 - 6, 6 - 7 + 1] = [3, 3, 1, 0] \\
\tau &= [0, 2, 4, 4, 3], \quad r = [0, 1, 4] \\
\tau[r] &= \tau[0, 1, 4] = [0, 2, 3] \\
Q[0] &= [2 - 0, 3 - 2 + 1] = [2, 2].
\end{aligned}$$

Comment:

$Q[2] = Q$ and $Q[i - 1]$ is the contraction vector of $Q[i]$ for $1 \leq i \leq 2$. The calculations of τ are done in Example 15.1 and 15.2. The calculations of r are done in Example 17.1 and 17.2. Moreover, the calculation of $Q[1]$ and $Q[0]$ are done in Example 18.1 and 18.2.

Step 0:

$j[0] = 2$ and $\zeta[0] = 5$ are the cyclic parameters of the extension $[2, 3]$ of $Q[0] = [2, 2]$. We refer to Example 20.3.

Step 1:

$$\begin{aligned}
Q[1] &= [3, 3, 1, 0] \\
C[1] &= [0, 3] \\
\tau &= [0, 2, 4, 4, 3] \\
D[1] &= [\tau(3)] = [4] \\
\gamma[1] &= 1, \quad \alpha[1] = 4, \quad \gamma^*[1] = 1, \quad \alpha^*[1] = 4 \\
g &= \gcd(\alpha^*[1], \zeta[0]) = \gcd(4, 5) = 1 \\
x &= \zeta[0]/g = 5/1 = 5, \quad y = \alpha^*[1]/g = 4/1 = 4 \\
j[1] &= 2 \cdot x \cdot \gamma^*[1] + y \cdot j[0] = 2 \cdot 5 \cdot 1 + 4 \cdot 2 = 18 \\
\zeta[1] &= y \cdot \zeta[0] + j[1] = 4 \cdot 5 + 18 = 38.
\end{aligned}$$

Comment: τ , $C[1]$ and $D[1]$ are calculated in Example 15.2 and 22.1. Moreover, $\gamma[1] = \#D[1] = 1$ and $\alpha[1] = \delta(Q[1]) + 1 = 4$. Since $\gamma[1] = 1$, then Observation 23.2 b) implies that $\alpha^*[1] = \alpha[1] = 4$ and $\gamma^*[1] = \gamma[1] = 1$. For the last calculations we refer to Case 2 in Section 7.

Step 2:

$$Q[2] = [4, 4, 2, 0], \quad j[2] = j[1] = 18, \quad \zeta[2] = \zeta[1] + j[2] = 38 + 18 = 56.$$

This is true since all coordinates except the last one are greater than 1. We refer to Case 1 in Section 7.

Example 8.2 - continued.

Start values: $Q = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1]$ and $p = 1$.

Since the first coordinate is larger than p , then Observation 11.3 implies that p is an admissible p -value of Q .

Contraction results:

$$Q[1] = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1]$$

$$\tau = [0, 1, 1, 2, 2, 2, 4, 4, 5, 5, 5, 7, 7, 7, 8, 8, 8], \quad r = [0, 5, 10, 13, 16]$$

$$\tau[r] = \tau[0, 5, 10, 13, 16] = [0, 2, 5, 7, 8]$$

$$Q[0] = [2 - 0, 5 - 2, 7 - 5, 8 - 7 + 1] = [2, 3, 2, 2].$$

Comment: $Q[1] = Q$ and $Q[0]$ is the contraction vector of $Q[1]$. The calculations of τ , r and $Q[0]$ are done in Example 15.3, 17.3 and 18.3.

Step 0: $j[0] = 2$ and $\zeta[0] = 5$ are the cyclic parameters of the extension $[2, 3, 2, 3]$ of $Q[0] = [2, 3, 2, 2]$. We refer to Example 20.4.

Step 1:

$$Q[1] = [2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1]$$

$$C[1] = [0, 2, 4, 7, 9, 12, 15]$$

$$\tau = [0, 1, 1, 2, 2, 2, 4, 4, 5, 5, 5, 7, 7, 7, 8, 8, 8]$$

$$D[1] = [\tau(2), \tau(4), \tau(7), \tau(9), \tau(12), \tau(15)] = [1, 2, 4, 5, 7, 8]$$

$$\gamma[1] = 6, \quad \alpha[1] = 9, \quad \gamma^*[1] = 2, \quad \alpha^*[1] = 3$$

$$g = \gcd(\alpha^*[1], \zeta[0]) = \gcd(3, 5) = 1$$

$$x = \zeta[0]/g = 5/1 = 5, \quad y = \alpha^*[1]/g = 3/1 = 3$$

$$j[1] = 2 \cdot x \cdot \gamma^*[1] + y \cdot j[0] = 2 \cdot 5 \cdot 2 + 3 \cdot 2 = 26$$

$$\zeta[1] = y \cdot \zeta[0] + j[1] = 3 \cdot 5 + 26 = 41.$$

Comment: τ , $C[1]$ and $D[1]$ are calculated in Example 15.3 and 22.2. Moreover, $\gamma[1] = \#D[1] = 6$ and $\alpha[1] = \delta(Q[1]) + 1 = 9$. By Example 23.5 we get that $\alpha^*[1] = 3$ and $\gamma^*[1] = 2$ are the least progression parameters of $D[1]$ with respect to $\alpha[1]$. For the last calculations we refer to Case 2 in Section 7.

Example 8.3 - continued.

Start values: $Q = [3, 3, 3, 3, 2, 2, 1, 998]$ and $p = 2$. Since the first coordinate is larger than p , then Observation 11.3 implies that p is an admissible p -value of Q .

Contraction results:

$$Q[2] = [3, 3, 3, 3, 2, 2, 1, 998]$$

$$\tau = [0, 2, 4, 6, 8, 9, 10, 1007], \quad r = [0, 1, 2, 3, 4, 5, 8]$$

$$\tau[r] = \tau[0, 1, 2, 3, 4, 5, 8] = [0, 2, 4, 6, 8, 9, 1007]$$

$$Q[1] = (2 - 0, 4 - 2, 6 - 4, 8 - 6, 9 - 8, 1007 - 9 + 1) = [2, 2, 2, 2, 1, 999]$$

$$\tau = [0, 1, 2, 3, 4, 4, 1002], \quad r = [0, 1, 2, 3, 6]$$

$$\tau[r] = \tau[0, 1, 2, 3, 6] = [0, 1, 2, 3, 1002]$$

$$Q[0] = (1 - 0, 2 - 1, 3 - 2, 1002 - 3 + 1) = [1, 1, 1, 1000].$$

Comment:

$Q[2] = Q$ and $Q[i - 1]$ is the contraction vector of $Q[i]$ for $1 \leq i \leq 2$. The calculations of τ are done in Example 15.4 and 15.5. The calculations of r are done in Example 17.4 and 17.5. Moreover, the calculation of $Q[1]$ and $Q[0]$ are done in Example 18.4 and 18.5.

Step 0: $j[0] = 4$ and $\zeta[0] = 1004$ are the cyclic parameters of the extension given by $[1, 1, 1, 1001]$ of $Q[0] = [1, 1, 1, 1000]$. We refer to Example 20.2.

Step 1:

$$Q[1] = [2, 2, 2, 2, 1, 999]$$

$$C[1] = [0, 5]$$

$$\tau = [0, 1, 2, 3, 4, 4, 1002]$$

$$D[1] = [\tau(5)] = [4]$$

$$\gamma[1] = 1, \quad \alpha[1] = 1003, \quad \gamma^*[1] = 1, \quad \alpha^*[1] = 1003$$

$$g = \gcd(\alpha^*[1], \zeta[0]) = \gcd(1003, 1004) = 1$$

$$x = \zeta[0]/g = 1004/1 = 1004, \quad y = \alpha^*[1]/g = 1003/1 = 1003$$

$$j[1] = 2 \cdot x \cdot \gamma^*[1] + y \cdot j[0] = 2 \cdot 1004 \cdot 1 + 1003 \cdot 4 = 6020$$

$$\zeta[1] = y \cdot \zeta[0] + j[1] = 1003 \cdot 1004 + 6020 = 1013032.$$

Comment: τ , $C[1]$ and $D[1]$ are calculated in Example 15.5 and 22.4. Moreover, $\gamma[1] = \#D[1] = 1$ and $\alpha[1] = \delta(Q[1]) + 1 = 1003$. Since $\gamma[1] = 1$, then Observation 23.2 b) implies that $\alpha^*[1] = \alpha[1] = 1003$ and $\gamma^*[1] = \gamma[1] = 1$. For the last calculations we refer to Case 2 in Section 7.

Step 2:

$$Q[2] = [3, 3, 3, 3, 2, 2, 1, 998]$$

$$C[2] = [0, 7]$$

$$\tau = [0, 2, 4, 6, 8, 9, 10, 10, 1007]$$

$$D[2] = [\tau(7)] = [10]$$

$$\gamma[2] = 1, \alpha[2] = 1008, \gamma^*[2] = 1, \alpha^*[2] = 1008$$

$$g = \gcd(\alpha^*[2], \zeta[1]) = \gcd(1008, 1013032) = 8$$

$$x = \zeta[1]/g = 1013032/8 = 126629, y = \alpha^*[2]/g = 1008/8 = 126$$

$$j[2] = 2 \cdot x \cdot \gamma^*[2] + y \cdot j[1] = 2 \cdot 126629 \cdot 1 + 126 \cdot 6020 = 1011778$$

$$\zeta[2] = y \cdot \zeta[1] + j[2] = 126 \cdot 1013032 + 1011778 = 128653810.$$

Comment: τ , $C[2]$ and $D[2]$ are calculated in Example 15.4 and 22.3. Moreover, $\gamma[2] = \#D[2] = 1$ and $\alpha[2] = \delta(Q[2]) + 1 = 1008$. Since $\gamma[2] = 1$, then Observation 23.2 b) implies that $\alpha^*[2] = \alpha[2] = 1008$ and $\gamma^*[2] = \gamma[2] = 1$. For the last calculations we refer to Case 2 in Section 7.